Polynomial decay for the Korteweg–de Vries equation at length 14π

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Abstract

We prove uniform small-data polynomial decay for the real Korteweg–de Vries equation on an interval of length 14π, with homogeneous Dirichlet boundary conditions and zero Neumann datum at the right endpoint. This critical length has three undamped linear modes, including a zero frequency. Every sufficiently small real L^2 energy solution satisfies $\|y(t)\|_2 \le C(1+t)^{-1/2}$, with constants independent of the datum. The quadratic response is made periodic only on the exponentially stable subspace. Resonant terms in the undamped subspace may grow, but their boundary observation vanishes. An explicit nonzero double-frequency observation then gives a uniform quartic energy loss, which yields the all-time decay estimate.

1 Introduction

The Korteweg–de Vries equation on a bounded interval is a basic example in the boundary control of dispersive equations. Homogeneous boundary conditions can dissipate its L^2 energy, even though the equation has no interior damping. The obstruction to linear stabilization is particularly concrete: at certain interval lengths a finite-dimensional space of linear solutions has zero boundary loss. These critical lengths, identified in Rosier's controllability analysis [7], are

$$\mathcal{N} = \left\{ 2\pi\sqrt{\frac{k^2+k\ell+\ell^2}{3}} : k,\ell \in \mathbb{N}^* \right\}.$$

Small nonlinear solutions are globally well posed in the usual energy class; we use the realization of Coron–Crépeau [2] and the global statement in [5, Lemma 3.2]. The issue here is a quantitative decay rate for that flow when linear boundary loss misses some directions.

Nguyen [4, Theorem 1.1] obtains decay at least of order $t^{-1/2}$ when the undamped space has dimension one, and for certain even-dimensional undamped spaces satisfying a nondegeneracy condition. His argument expands the flow over a fixed time interval and detects a quadratic boundary response. A zero critical frequency interacting with a nonzero one can prevent the full quadratically forced solution from having a bounded separated form [4, Remark 2.6]. The case of odd undamped dimension greater than one lies outside those decay theorems. Polynomial decay in that case is explicitly left open in [5, Section 2.4.5]. Recent estimates near critical lengths [6, Theorem 1.6] exclude the exact critical value in that theorem and do not settle the present question.

We consider the concrete length $L = 14\pi$. Its critical pairs are $(7,7), (2,11), (11,2)$, giving one stationary mode and a conjugate oscillatory pair. For the real equation

$$\begin{cases} y_t + y_x + y_{xxx} + yy_x = 0, & t>0, \quad 0<x<L, \\ y(t,0) = y(t,L) = y_x(t,L) = 0, & t>0, \\ y(0,x) = y_0(x), & 0<x<L, \end{cases} \tag{1.1}$$

we prove the following uniform bound. An energy solution means the usual mild solution in $C([0,\infty);L^2(0,L))\cap L^2_{\mathrm{loc}}([0,\infty);H^1_0(0,L))$; Neumann boundary values are interpreted through the standard L^2-in-time hidden trace, recalled in Section 2.

Theorem 1.1. *Let $L=14\pi$. There exist $\delta,C>0$ such that every real energy solution of (1.1) with $\|y_0\|_{L^2(0,L)}\le\delta$ satisfies*

$$\|y(t)\|_{L^2(0,L)}\le C(1+t)^{-1/2}\qquad \text{for every } t\ge 0. \tag{1.2}$$

The constants are chosen independently of the initial datum.

This supplies polynomial decay at an exact critical length with a three-dimensional undamped space. The estimate allows every sufficiently small real L^2 datum, including arbitrary mixtures of the stationary, oscillatory and stable components. In particular it gives an affirmative answer at 14π to the odd-dimensional decay question above. We do not claim the same conclusion for all critical lengths or an optimal rate.

The mixed resonance is dealt with by retaining the entire center forcing through Duhamel's formula. Only the stable part of the quadratic response is chosen periodic. The remaining center motion can have secular growth, but its boundary derivative is identically zero. The observation is therefore a finite temporal Fourier sum. Its constant term detects the stationary mode; an explicit double-frequency coefficient detects the oscillatory pair. Orthogonality over one period gives a positive lower bound uniformly over all real center directions. A fixed late window removes the stable homogeneous transient. The quadratic expansion has a cubic remainder in the boundary observation norm for rough L^2 data, giving a quartic energy loss on a single, datum-independent interval. Iterating that loss proves (1.2). The fixed-window comparison follows Nguyen's approach [4, Section 5]; allowing the invisible resonant response is the change needed here.

Section 2 determines the undamped subspace and proves exponential decay on its orthogonal complement. Section 3 constructs the quadratic observation and proves its uniform lower bound. Section 4 establishes the nonlinear energy loss and proves the theorem. All spatial norms below are on $(0,L)$; complex notation is used only for linear calculations, and $\langle f,g\rangle=\int f\overline{g}$.

2 Linear evolution and its undamped subspace

Put $L=14\pi$, and write $H=L^2(0,L;\mathbb{C})$ with inner product $\langle f,g\rangle=\int_0^L f\overline{g}\,dx$, linear in its first argument. The real subspace will be used for the nonlinear equation. Let

$$Av=-v'-v''',\qquad D(A)=\{v\in H^3(0,L):v(0)=v(L)=v'(L)=0\}. \tag{2.1}$$

Rosier's Proposition 3.1 [7] gives a contraction semigroup $S(t)$ on the real L^2 space. Extending it by real and imaginary parts gives the complex contraction semigroup with precisely the domain in (2.1). On $D(A)$ define $\mathcal{B}v=v'(0)$. Integration by parts gives

$$\mathrm{Re}\langle Av,v\rangle=-\tfrac12|\mathcal{B}v|^2. \tag{2.2}$$

The adjoint is $A^*v=v'+v'''$ on $\{v\in H^3:v(0)=v(L)=v'(0)=0\}$. The semigroup difference quotient shows that $S(t)$ preserves $D(A)$, commutes there with A, and evolves an eigenvector $Av=\lambda v$ as $S(t)v=e^{\lambda t}v$.

2.1 Mild solutions and boundary estimates

For a fixed $T > 0$ set

$$X_T = C([0,T];H) \cap L^2(0,T;H_0^1(0,L)), \qquad \|v\|_{X_T} = \|v\|_{C_tH} + \|v\|_{L^2_tH^1}.$$

Proposition 7 of Coron–Crépeau [2] provides continuous linear maps from $H \times L^1(0,T;H)$ to the mild solution $v \in X_T$ of $v_t = Av + f$, $v(0) = h$, and to its left derivative in $L^2(0,T)$, with zero right Neumann datum. The latter map defines $\mathcal{B}v$ for rough data. It agrees with the actual derivative for classical solutions. In particular,

$$\|v\|_{X_T} + \|\mathcal{B}v\|_{L^2(0,T)} \le K_T(\|h\|_2 + \|f\|_{L^1_tH}). \tag{2.3}$$

Here and below $\|\cdot\|_2$ is the spatial L^2 norm. The identities behind this estimate are

$$\|v(t)\|_2^2 + \int_0^t |\mathcal{B}v(s)|^2\,ds = \|h\|_2^2 + 2\,\mathrm{Re}\int_0^t \langle f(s), v(s)\rangle\,ds, \tag{2.4}$$

$$\frac{d}{dt}\int_0^L x|v|^2\,dx + 3\|v_x\|_2^2 = \|v\|_2^2 + 2\,\mathrm{Re}\int_0^L xf\overline{v}\,dx. \tag{2.5}$$

For smooth data these follow by integration by parts. The first gives $\sup_t \|v(t)\|_2 \le \|h\|_2 + \|f\|_{L^1_tH}$ and the trace bound; the integrated second identity gives the spatial derivative bound. To pass to mild data, approximate h in H by $C_c^\infty(0,L)$ and f in L^1_tH by finite sums of smooth time functions with spatial values in $C_c^\infty(0,L)$. The continuous maps of [2, Proposition 7] give strong convergence in C_tH, $L^2_tH^1$ and L^2 for the left trace. Every term in the integrated identities converges, uniformly in its upper endpoint. Zero Dirichlet traces pass in H^1, so the limit belongs to X_T. Thus (2.4) holds for every $t \in [0,T]$, and (2.5) holds in its integrated form and almost everywhere. This also identifies the mild solution with $S(t)h + \int_0^t S(t-s)f(s)\,ds$. The hidden trace respects time shifts: prove $\mathcal{B}S(s)S(t)h = \mathcal{B}S(s+t)h$ first for $h \in D(A)$ and then use its continuous extension in $L^2(0,T)$. We shall use this identity for fixed $t \ge 0$, without assigning a boundary derivative to an arbitrary state in H.

We also need graph-domain control of this observation. If $f = -Av = v''' + v'$, then, with $a = v'(0)$ and $b = v''(0)$,

$$v(x) = a\sin x + b(1-\cos x) + F(x), \qquad F(x) = \int_0^x (1-\cos(x-s))f(s)\,ds.$$

This formula has the prescribed three initial derivatives and satisfies the ODE. Uniqueness with zero initial derivatives follows by differentiating $|v'|^2 + |v''|^2$ for the homogeneous equation. The convolution kernels bound F, F', F'' in L^2, and $F''' = f - F'$. The Gram matrix of $\sin x$ and $1-\cos x$ is invertible, hence their coefficients are bounded by $C_L(\|v\|_2 + \|F\|_2)$. Consequently

$$\|v\|_{H^3} + |\mathcal{B}v| \le C_L(\|Av\|_2 + \|v\|_2), \qquad v \in D(A). \tag{2.6}$$

All endpoint values here use the continuous one-dimensional Sobolev representatives; the corresponding Sobolev facts may be found in [1, Theorems 8.2, 8.8 and 8.12].

2.2 The three undamped modes

The following explicit spectral description will also identify the subspace on which linear decay fails.

Lemma 2.1. *The only imaginary eigenvalues of A are $0, \pm ip$, where*

$$p = \frac{120}{343}, \qquad \phi(x) = 1 - \cos x, \qquad \psi(x) = 2e^{-8ix/7} - 13e^{3ix/7} + 11e^{5ix/7}. \tag{2.7}$$

Their eigenspaces span $M = \operatorname{span}_{\mathbb{C}}\{\phi, \psi, \overline{\psi}\}$. They are mutually orthogonal and have zero derivative at both endpoints. Every real trajectory in M is

$$U(t) = c\phi + ze^{-ipt}\psi + \overline{z}e^{ipt}\overline{\psi}, \qquad c \in \mathbb{R}, \quad z \in \mathbb{C}, \tag{2.8}$$

with

$$\|U(0)\|_2^2 = \frac{3L}{2}c^2 + 588L|z|^2. \tag{2.9}$$

Proof. If $Av = -i\omega v$, $\omega \in \mathbb{R}$, then (2.2) also gives $v'(0) = 0$. The spatial roots satisfy $\lambda^3 + \lambda - i\omega = 0$. For distinct roots, the coefficients of $v = \sum a_j e^{\lambda_j x}$ and of its endpoint values both lie in the one-dimensional nullspace of the rows $(1, 1, 1)$ and $(\lambda_1, \lambda_2, \lambda_3)$. Each coordinate of its null vector is nonzero. Thus $e^{\lambda_1 L} = e^{\lambda_2 L} = e^{\lambda_3 L}$. Their real parts must be equal, and their sum is zero, so all roots are purely imaginary. At a repeated root the roots are $ir, ir, -2ir$, with $r = \pm 1/\sqrt{3}$. The zero initial value and derivative give $v = ae^{-2irx} + (-a + 3irax)e^{irx}$. If $a \neq 0$, its endpoint value requires $e^{-3irL} = 1 - 3irL$, impossible by absolute values. The remaining solution is zero. These exponential descriptions exhaust the ODE solutions: the initial-jet matrix is a Vandermonde matrix (or its double-root version), and a solution with all initial jets zero vanishes by the differential inequality $\frac{d}{dx}|(v, v', v'')|^2 \leq (1 + |\omega|)|(v, v', v'')|^2$.

Write the three distinct roots as iq_1, iq_2, iq_3 in increasing order. The common endpoint phase implies $q_2 - q_1 = 2\pi k/L$, $q_3 - q_2 = 2\pi\ell/L$ for positive integers k, ℓ. Viète's relations give $L = 2\pi\sqrt{(k^2 + k\ell + \ell^2)/3}$. At $L = 14\pi$ this is $k^2 + k\ell + \ell^2 = 147$. Here $k \leq 12$, and the discriminants for $k = 1, \ldots, 12$ are

$$585, 576, 561, 540, 513, 480, 441, 396, 345, 288, 225, 156.$$

Only $576 = 24^2$, $441 = 21^2$ and $225 = 15^2$ are squares, giving $(k, \ell) = (2, 11), (7, 7), (11, 2)$. Each eigenspace has dimension one by the same endpoint nullspace. Differentiating (2.7) gives $A\phi = 0$ and $A\psi = -ip\psi$, with all four endpoint values zero. Finally every nonconstant exponential of frequency an integer multiple of $1/7$ integrates to zero over $(0, 14\pi)$. This proves the orthogonality and (2.9), since $2^2 + 13^2 + 11^2 = 294$. □

2.3 Decay on the orthogonal complement

Let P be the orthogonal projection onto M and $Q = I - P$.

Lemma 2.2. *The spaces M and $M^{\perp}$ reduce $S(t)$. There are $K_s \geq 1$ and $\gamma > 0$, depending only on L, such that*

$$\|S(t)Q\|_{H \to H} \leq K_s e^{-\gamma t} \qquad (t \geq 0). \tag{2.10}$$

Moreover Q preserves $D(A)$ and $AQ = QA$ there.

Proof. For $T > 0$ put $N_T = \{h \in H : \mathcal{B}S(\cdot)h = 0 \text{ in } L^2(0, T)\}$. First, bounded initial states give a relatively compact family $S(\cdot)h$ in $L^2(0, T; H)$. Indeed (2.3) bounds these trajectories in $L^2_t H^1_0$. For each $a > 0$ compact spatial embedding and smooth density give a finite-dimensional subspace $E_a \subset C_c^{\infty}(0, L)$ with $\|u - P_{E_a}u\|_2 \leq a\|u\|_{H^1}$ for every $u \in H^1_0$. For a smooth basis test φ of this space the coefficient $b(t) = \langle S(t)h, \varphi\rangle$ satisfies $b' = -\langle (S(t)h)_x, \varphi + \varphi''\rangle$ in distributions. It is

bounded in $H^1(0,T)$ and hence compact in $C([0,T])$. The finitely many coefficients are therefore compact, while the spatial remainder is at most $a\|S(\cdot)h\|_{L^2_t H^1}$. A diagonal subsequence as $a \downarrow 0$ proves the assertion.

The kernel N_T is closed by trace continuity. On this kernel (2.4) makes $h \mapsto S(\cdot)h$ an isometry up to $\sqrt{T}$. Thus its unit ball is compact and $\dim N_T < \infty$. The nonincreasing integer sequence $\dim N_n$ eventually stabilizes. The nested subspaces $N_n \subset N_1$ therefore stabilize, so for some integer T_0,

$$N_{T_0} = \bigcap_{T>0} N_T =: E.$$

Trace shifts give $S(t)E \subset E$. On this finite-dimensional space $S(t)$ is an isometry, hence onto. We must also know that $E \subset D(A)$. The averaged operators $T_h = h^{-1}\int_0^h S(s)|_E\, ds$ converge to I in operator norm, because strong continuity is uniform in finite dimension. They are invertible for small h, and the semigroup difference quotient gives $T_h v \in D(A)$ and $AT_h v = (S(h)v - v)/h$. Surjectivity yields the desired domain inclusion. The real generator $D = A|_{E\cap H_{\mathbb{R}}}$ is skew-adjoint by polarization of the norm identity. The self-adjoint nonnegative matrix $-D^2$ has an orthonormal eigenbasis. On a positive eigenspace with eigenvalue λ, a unit vector u and $Du/\sqrt{\lambda}$ form a rotation plane; its orthogonal complement is D-invariant. The zero eigenspace is the kernel of D. Thus the complexification of E has imaginary eigenvectors of A, and Lemma 2.1 implies $E \subset M$. The reverse inclusion follows from the zero observations of the explicit modes, so $E = M$.

There is consequently a $\theta > 0$ such that

$$\|\mathcal{B}S(\cdot)h\|_{L^2(0,T_0)} \geq \theta\|h\|_2 \qquad (h \in M^\perp). \tag{2.11}$$

Otherwise take unit $h_n \in M^\perp$ whose observations tend to zero. By the compactness just proved their trajectories have a Cauchy subsequence. Applying the time-averaged energy identity to $h_n - h_m$ gives

$$\|h_n - h_m\|_2^2 \leq T_0^{-1}\|S(\cdot)(h_n - h_m)\|^2_{L^2_t H} + \|\mathcal{B}S(\cdot)(h_n - h_m)\|^2_{L^2(0,T_0)}.$$

The initial states converge to a unit vector in $M^\perp \cap N_{T_0}$, which is impossible. Because $S(t)$ is contractive and unitary on M, applying contractivity to $m + ah$ for every $a \in \mathbb{C}$, $m \in M$, $h \in M^\perp$, forces $\langle S(t)m, S(t)h\rangle = 0$. This proves the reducing property. Decrease θ if necessary to $0 < \theta \leq 1/2$. Energy and (2.11) give $\|S(T_0)Q\| \leq q := \sqrt{1-\theta^2} < 1$. Iteration and contractivity between these times give (2.10), with $K_s = q^{-1}$ and $\gamma = -\log q/T_0$. Commutation of P, Q with $S(t)$ and the generator difference quotient prove the last assertion. □

3 A coercive quadratic boundary observation

The linear trajectories in M lose no energy. Their quadratic forcing must therefore produce the first observable loss. We construct a response whose boundary derivative is periodic, while allowing its unobserved part to grow.

3.1 Separating the stable response from the center motion

Recall the real trajectory $U(t) = c\phi + ze^{-ipt}\psi + \overline{z}e^{ipt}\overline{\psi}$ from (2.8), with $c \in \mathbb{R}$ and $z \in \mathbb{C}$. Put

$$F = -UU_x = \sum_{j=-2}^{2} e^{-ijpt}F_j. \tag{3.1}$$

The coefficients are smooth functions, quadratic in $c, z, \overline{z}$. For $j = -2, \dots, 2$ define

$$R_j f = \int_0^\infty e^{ijpt} S(t) Q f \, dt. \tag{3.2}$$

The stable estimate makes this a norm-convergent integral in QH, bounded by $K_s \gamma^{-1} \|Qf\|_2$. Its graph-domain meaning follows directly from

$$S(h) R_j f = e^{-ijph} \left(R_j f - \int_0^h e^{ijps} S(s) Q f \, ds \right).$$

Taking the right difference quotient proves

$$R_j f \in QD(A), \qquad (-ijp - A) R_j f = Qf. \tag{3.3}$$

This inverse is unique on QH, since a nonzero homogeneous solution would contradict (2.10). Estimate (2.6) bounds its boundary trace, and $R_{-j}(\overline{f}) = \overline{R_j f}$.

Set

$$W_s(t) = \sum_{j=-2}^{2} e^{-ijpt} R_j F_j, \qquad W_c(t) = \int_0^t S(t-s) P F(s) \, ds, \qquad W = W_s + W_c. \tag{3.4}$$

These graph-domain functions solve $W_t = AW + F$. Although W_c can have secular terms, it lies in M and $\mathcal{B} W_c = 0$. Therefore

$$G(t) = \mathcal{B} W(t) = \sum_{j=-2}^{2} e^{-ijpt} \mathcal{B} R_j F_j \tag{3.5}$$

is periodic with period $P_0 = 2\pi/p$. Moreover $W(0) = W_s(0)$ is bounded in H, uniformly for $\|U(0)\|_2 \le 1$. Only this initial bound, rather than an all-time bound on W_c, will be needed.

3.2 The constant and double-frequency coefficients

The zero-mode self-interaction has a stationary response

$$w_{00}(x) = \tfrac{1}{2}(x - L) \sin x + \tfrac{1}{12}(\cos 2x - 1). \tag{3.6}$$

It belongs to $D(A)$ and satisfies $w_{00}''' + w_{00}' = -\phi\phi'$, $\mathcal{B} w_{00} = -L/2$. Its forcing is orthogonal to M: integer frequencies do not meet those of $\psi, \overline{\psi}$, and $\int \phi^2 \phi' = 0$. Thus $Q w_{00}$ is its stable response and has the same boundary derivative.

The oscillatory mean forcing is $-|z|^2 (|\psi|^2)'$, where

$$|\psi|^2 = 294 - 52 \cos(11x/7) + 44 \cos(13x/7) - 286 \cos(2x/7). \tag{3.7}$$

The differentiated frequencies $\pm 2/7, \pm 11/7, \pm 13/7$ are disjoint from those of every element of M, so this forcing is orthogonal to M. Writing $|\psi|^2 = \sum d_q e^{iqx}$, the function

$$h(x) = \sum_{q \neq 0} \frac{-d_q}{1 - q^2} (e^{iqx} - 1)$$

satisfies $h''' + h' = -(|\psi|^2)'$ and all three domain boundary conditions. Indeed $qL \in 2\pi\mathbb{Z}$, and $d_q = d_{-q} \in \mathbb{R}$ makes its endpoint derivative sum vanish. Hence $\mathcal{B} h = 0$, and projection onto QH does not change this observation. The constant term of G is exactly $-Lc^2/2$.

At frequency $2p$, let v_2 be the solution of

$$v_2''' + v_2' - 2ipv_2 = -\psi\psi', \qquad v_2(0) = v_2(L) = v_2'(L) = 0, \tag{3.8}$$

and set $\beta_2 = \mathcal{B}v_2$. Existence and uniqueness follow by combining (3.3) on $M^\perp$ with the invertible center block, whose factors are $-2ip, -ip, -3ip$. Its full solution and its stable component have the same observation. Also put $\beta_1 = \mathcal{B}R_1[-(\phi\psi)']$, a finite number by (2.6). Expansion of (3.1) now gives

$$G(t) = -\tfrac{L}{2}c^2 + cz\beta_1 e^{-ipt} + c\overline{z}\,\overline{\beta_1}e^{ipt} + z^2\beta_2 e^{-2ipt} + \overline{z}^2\overline{\beta_2}e^{2ipt}. \tag{3.9}$$

We need no nonvanishing assertion for β_1.

Lemma 3.1. *The coefficient β_2 is nonzero.*

Proof. Write $q = (-8/7, 3/7, 5/7)$ and $a = (2, -13, 11)$. Since $q_j - q_j^3 = p$, a particular solution of (3.8) is

$$\chi(x) = \frac{1}{6}\left(\sum_{j=1}^{3}\frac{a_j}{q_j}e^{iq_jx}\right)^2.$$

For verification, if $\xi = q_j + q_k$ then $\xi - \xi^3 - 2p = -3q_jq_k(q_j + q_k)$; applying the differential operator to the ordered quadratic sum gives precisely $-\psi\psi'$. At both endpoints $\chi = d$ and $\chi' = 0$, with

$$d = \frac{1}{6}\left(-\frac{7}{4} - \frac{91}{3} + \frac{77}{5}\right)^2 = \frac{1002001}{21600} > 0. \tag{3.10}$$

Let $\rho > 1$ solve $\rho^3 - \rho = 240/343$. Monotonicity and rational substitution give

$$\frac{31}{25} < \rho < \frac{5}{4}. \tag{3.11}$$

The spatial roots of (3.8) are $\lambda_0 = -i\rho$ and $\lambda_\pm = \pm\sigma + i\rho/2$, where $\sigma = \sqrt{3\rho^2/4 - 1} > 0$. Put $E_j = e^{\lambda_j L}$; their product is one. The homogeneous correction $v_2 - \chi = \sum b_j e^{\lambda_j x}$ satisfies

$$\mathsf{Q}b = (-d, -d, 0)^T, \qquad \mathsf{Q} = \begin{pmatrix} 1 & 1 & 1 \\ E_0 & E_+ & E_- \\ \lambda_0 E_0 & \lambda_+ E_+ & \lambda_- E_- \end{pmatrix}.$$

This matrix is invertible by the uniqueness already established for (3.8). Expanding its adjugate and using $E_0E_+E_- = 1$ gives

$$\beta_2 = -\frac{d}{\det \mathsf{Q}}\sum_{j=0}^{2}\lambda_j(\lambda_{j+2} - \lambda_{j+1})(E_j + E_j^{-1}), \tag{3.12}$$

where indices are cyclic in the order $(0, +, -)$. To examine the numerator, put $\vartheta = \rho L/2 = 7\pi\rho$, $v = \sigma L$ and $K = 3\rho^2/2 - 1$. The three cyclic prefactors in (3.12) are $2i\rho\sigma$, $K - i\rho\sigma$ and $-K - i\rho\sigma$. Since $E_0 + E_0^{-1} = 2\cos(2\vartheta)$ and $E_\pm + E_\pm^{-1} = 2\cosh(\pm v + i\vartheta)$, the sum is

$$4i\big[\rho\sigma\cos(2\vartheta) + K\sinh(v)\sin\vartheta - \rho\sigma\cosh(v)\cos\vartheta\big]. \tag{3.13}$$

By (3.11), ϑ modulo 2π is $\pi(7\rho - 8)$, with $17/25 < 7\rho - 8 < 3/4$. Thus $\sin\vartheta > 1/2$ and $\cos\vartheta < 0$. Furthermore $\sigma^2 > 383/2500$, so $v > 2$; $\sigma < 1/2$ and $\rho\sigma < 5/8$; and $K > 1633/1250 > 13/10$. The real bracket in (3.13) is therefore larger than

$$-\frac{5}{8} + \frac{13}{10}\cdot 1\cdot\frac{1}{2} = \frac{1}{40} > 0.$$

Thus (3.12) cannot vanish. □

Proposition 3.2. *There is an $\eta > 0$ such that every real center trajectory with $1/2 \le \|U(0)\|_2 \le 1$ satisfies*

$$\|G\|_{L^2(T,T+P_0)} \ge \eta \qquad \text{for every } T \ge 0. \tag{3.14}$$

Proof. Integrating (3.9) over a full period gives

$$\int_T^{T+P_0} |G(t)|^2\,dt = P_0\left(\frac{L^2}{4}c^4 + 2|\beta_1|^2c^2|z|^2 + 2|\beta_2|^2|z|^4\right).$$

The cross terms have distinct temporal frequencies, hence their integrals vanish for every starting time T. By (2.9), the lower shell bound gives $c^2+|z|^2 \ge 1/(4\cdot 588L)$. Since $c^4+|z|^4 \ge (c^2+|z|^2)^2/2$, one may choose

$$\eta^2 = \frac{P_0}{32(588L)^2}\min\{L^2/4, 2|\beta_2|^2\} > 0.$$

This one constant covers the zero mode, the oscillatory pair, and every mixed direction and phase. □

4 Uniform nonlinear energy loss and decay

We now compare a small nonlinear solution with the quadratic responses on one fixed time interval. All observation times will be chosen before the smallness threshold and the initial datum.

Small-data global well-posedness in the usual energy class is given by [5, Lemma 3.2] with zero Neumann control. The mild realization is the one in [2, Proposition 7 and Appendix A]: on each finite interval, $y = S(\cdot)y_0 - \int_0^{\cdot} S(\cdot - s)(yy_x)(s)\,ds$. The source's usual energy solution has precisely this meaning; see also [3, Lemma 5.4]. Only existence and uniqueness are used from the nonlinear well-posedness statement.

For such a real solution $yy_x \in L^1_t H$ on every finite interval, by $H^1 \hookrightarrow L^\infty$ and time Hölder. We may apply the linear identities with this exact force. Since $\int_0^L y^2 y_x\,dx = 0$ for $y \in H_0^1$, we obtain

$$E(t) + \int_0^t |\mathcal{B}y(s)|^2\,ds = E(0), \qquad E(t) = \|y(t)\|_2^2. \tag{4.1}$$

The weighted identity becomes

$$\frac{d}{dt}\int_0^L xy^2\,dx + 3\|y_x\|_2^2 = \|y\|_2^2 + \frac{2}{3}\int_0^L y^3\,dx.$$

These polynomial integration-by-parts identities hold for H_0^1 by its smooth density and the continuous $H^1 \to L^\infty$ embedding. If $\varepsilon = \|y_0\|_2 \le 1$, energy and $|\int y^3| \le \sqrt{L}\,\varepsilon^2\|y_x\|_2$, followed by Young's inequality, give for each fixed T

$$\|y\|_{X_T} \le D_T\varepsilon. \tag{4.2}$$

In particular this estimate requires no initial Sobolev regularity. The bilinear estimate

$$\|ab_x\|_{L^1(0,T;H)} \le C_L\|a\|_{L^2_tH^1}\|b\|_{L^2_tH^1} \tag{4.3}$$

and (2.3) yield

$$\|y - S(\cdot)y_0\|_{X_T} + \|\mathcal{B}(y - S(\cdot)y_0)\|_{L^2(0,T)} \le D'_T\varepsilon^2. \tag{4.4}$$

Proposition 4.1. *There exist $T_* > 0$, $\delta > 0$ and $\kappa > 0$, depending only on L, such that every real solution with $\|y_0\|_2 \le \delta$ satisfies*

$$E(T_*) \le E(0) - \kappa E(0)^2. \tag{4.5}$$

Proof. The zero datum is immediate from (4.1); assume $\varepsilon > 0$. Choose T_1 so large that energy and (2.10) give $\|\mathcal{B}S(\cdot)h\|_{L^2(0,T_1)} \geq \frac{1}{2}\|h\|_2$ for $h \in M^\perp$. Choose $\beta > 4D'_{T_1}$. The scale $\beta\varepsilon^2$ separates initial stable components already visible in the linear approximation from data whose first relevant boundary signal is quadratic.

If $\|Qy_0\|_2 \geq \beta\varepsilon^2$, then (4.4) gives

$$\|\mathcal{B}y\|_{L^2(0,T_1)} \geq \tfrac{1}{2}\|Qy_0\|_2 - D'_{T_1}\varepsilon^2 \geq \tfrac{\beta}{4}\varepsilon^2. \tag{4.6}$$

Here $\mathcal{B}S(t)Py_0 = 0$.

Suppose instead that $\|Qy_0\|_2 < \beta\varepsilon^2$, and put

$$U(t) = \varepsilon^{-1}S(t)Py_0, \qquad r = \varepsilon^{-2}Qy_0.$$

For $\varepsilon \leq 1/(2\beta)$, orthogonality gives $1/2 \leq \|U(0)\|_2 \leq 1$ and $\|r\|_2 \leq \beta$. Let $V_t = AV - UU_x$, $V(0) = r$. The response (3.4) gives

$$V(t) = W(t) + S(t)(r - W(0)). \tag{4.7}$$

Both r and $W(0)$ lie in $M^\perp$. Let A_0 bound $\|W(0)\|_2$ on the shell above. Choose a fixed T_2 such that $K_s e^{-\gamma T_2}(\beta + A_0) \leq \eta/2$. The boundary trace of the homogeneous term in (4.7) on $(T_2, T_2 + P_0)$ is at most its state norm at T_2, by the shifted energy identity. Proposition 3.2 therefore gives

$$\|\mathcal{B}V\|_{L^2(T_2,T_2+P_0)} \geq \eta/2. \tag{4.8}$$

Set $T_* = \max\{T_1, T_2 + P_0\}$. Equations (4.4) and (2.3) imply $\|y - \varepsilon U\|_{X_{T_*}} \leq C\varepsilon^2$, uniformly on this case. Indeed the omitted linear term is $S(t)Qy_0$ with initial norm at most $\beta\varepsilon^2$. Using (4.3) on $(y - \varepsilon U)y_x + \varepsilon U(y - \varepsilon U)_x$ gives

$$\|yy_x - \varepsilon^2 UU_x\|_{L^1(0,T_*;H)} \leq C\varepsilon^3.$$

The function $y - \varepsilon U - \varepsilon^2 V$ has zero initial value and this force with the opposite sign. Estimate (2.3) thus bounds its boundary trace by $D\varepsilon^3$, for one $D \geq 1$ fixed in advance. Since $\mathcal{B}U = 0$, (4.8) implies, if $\varepsilon \leq \eta/(4D)$,

$$\|\mathcal{B}y\|_{L^2(0,T_*)} \geq \eta\varepsilon^2/4. \tag{4.9}$$

The possible secular growth of W_c causes no additional estimate: this comparison uses a fixed interval, and its observation vanishes exactly.

Choose $\kappa = \min\{\beta^2/16, \eta^2/16\} > 0$. Choose δ smaller than the global well-posedness threshold, 1, $1/(2\beta)$, $\eta/(4D)$ and $1/\sqrt{2\kappa}$. Every constant and time has now been fixed independently of the initial datum. Squaring (4.6) or (4.9) and using (4.1) proves (4.5). □

Proof of Theorem 1.1. Energy keeps every shifted trajectory inside the same small ball, and the equation is autonomous. Thus $E_n = E(nT_*)$ satisfies $E_{n+1} \leq E_n(1 - \kappa E_n)$. If some $E_n = 0$, the solution subsequently vanishes by (4.1). Otherwise, since $\kappa\delta^2 \leq 1/2$,

$$\frac{1}{E_{n+1}} \geq \frac{1}{E_n} + \kappa, \qquad E_n \leq \frac{E_0}{1 + \kappa n E_0} \leq \frac{1}{\kappa n} \quad (n \geq 1).$$

For $t \geq 2T_*$, $n = \lfloor t/T_* \rfloor \geq t/(2T_*)$ and monotonicity gives $E(t) \leq 2T_*/(\kappa t)$. On $[0, 2T_*]$ we have $E(t) \leq \delta^2$. Hence the theorem holds with

$$C = \max\{\delta\sqrt{1 + 2T_*}, \sqrt{(2T_* + 1)/\kappa}\}.$$

This includes zero data, arbitrarily small nonzero data, and every time between the observation windows. □

AI-use statement

Substantial OpenAI Codex assistance was used in mathematical exploration, calculations, source checking and preparation of this manuscript. Two additional fresh-context Codex agents supplied reader and referee reports, followed by an author revision. All three agents used the same vendor and shared host filesystem. Independent mathematical review is requested.

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